

A Practical Guide to Linear discriminant analysis

TOPIC -- LINEAR DISCRIMINANT ANALYSIS

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Discrimination rules Maximum likelihood: Assigns x to the group that maximizes population (group) density. Bayes Discriminant Rule: Assigns x to the group that maximizes $\pi_i f_i(x)$, where π_i represents the prior probability of that classification, and $f_i(x)$ represents the population density. Fisher's linear discriminant rule: Maximizes the ratio between SS_{between} and SS_{within} , and finds a linear combination of the predictors to predict group.

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Linear discriminant in high dimensions Geometric anomalies in higher dimensions lead to the well-known curse of dimensionality. Nevertheless, proper utilization of concentration of measure phenomena can make computation easier. An important case of these blessing of dimensionality phenomena was highlighted by Donoho and Tanner: if a sample is essentially high-dimensional then each point can be separated from the rest of the sample by linear inequality, with high probability, even for exponentially large samples. These linear inequalities can be selected in the standard (Fisher's) form of the linear discriminant for a rich family of probability distribution. In particular, such theorems are proven for log-concave distributions including multidimensional normal distribution (the proof is based on the concentration inequalities for log-concave measures) and for product measures on a multidimensional cube (this is proven using Talagrand's concentration inequality for product probability spaces). Data separability by classical linear discriminants simplifies the problem of error correction for artificial intelligence systems in high dimension.

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Structure Correlation Coefficients: The correlation between each predictor and the discriminant score of each function. This is a zero-order correlation (i.e., not corrected for the other predictors). Standardized Coefficients: Each predictor's weight in the linear combination that is the discriminant function. Like in a regression equation, these coefficients are partial (i.e., corrected for the other predictors). Indicates the unique contribution of each predictor in predicting group assignment. Functions at Group Centroids: Mean discriminant scores for each grouping variable are given for each function. The farther apart the means

are, the less error there will be in classification.

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Discriminant functions Discriminant analysis works by creating one or more linear combinations of predictors, creating a new latent variable for each function. These functions are called discriminant functions. The number of functions possible is either $N_g - 1$ where N_g = number of groups, or p (the number of predictors), whichever is smaller. The first function created maximizes the differences between groups on that function. The second function maximizes differences on that function, but also must not be correlated with the previous function. This continues with subsequent functions with the requirement that the new function not be correlated with any of the previous functions. Given group j , with R_j sets of sample space, there is a discriminant rule such that if $x \in R_j$, then $x \in j$. Discriminant analysis then, finds "good" regions of R_j to minimize classification error, therefore leading to a high percent correct classified in the classification table. Each function is given a discriminant score to determine how well it predicts group placement.

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Linear discriminant analysis (LDA), normal discriminant analysis (NDA), canonical variates analysis (CVA), or discriminant function analysis is a generalization of Fisher's linear discriminant, a method used in statistics and other fields, to find a linear combination of features that characterizes or separates two or more classes of objects or events. The resulting combination may be used as a linear classifier, or, more commonly, for dimensionality reduction before later classification. LDA is closely related to analysis of variance (ANOVA) and regression analysis, which also attempt to express one dependent variable as a linear combination of other features or measurements. However, ANOVA uses categorical independent variables and a continuous dependent variable, whereas discriminant analysis has continuous independent variables and a categorical dependent variable (i.e. the class label). Logistic regression and probit regression are more similar to LDA than ANOVA is, as they also explain a categorical variable by the values of continuous independent variables. These other methods are preferable in applications where it is not reasonable to assume that the independent variables have a normal distribution, which is a fundamental assumption of the LDA method. LDA is also closely related to principal component analysis (PCA) and factor analysis in that they both look for

linear combinations of variables which best explain the data. LDA explicitly attempts to model the difference between the classes of data. PCA, in contrast, does not take into account any difference in class, and factor analysis builds the feature combinations based on similarities rather than differences. Discriminant analysis is also different from factor analysis in that it is not an interdependence technique: a distinction between independent variables and dependent variables (also called criterion variables) must be made. LDA works when the measurements made on independent variables for each observation are continuous quantities. When dealing with categorical independent variables, the equivalent technique is discriminant correspondence analysis. Discriminant analysis is used when groups are known a priori (unlike in cluster analysis). Each case must have a score on one or more quantitative predictor measures, and a score on a group measure. In simple terms, discriminant function analysis is classification - the act of distributing things into groups, classes or categories of the same type.

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$$\vec{w} \rightarrow \propto (\Sigma_0 + \Sigma_1)^{-1} (\mu_0 - \mu_1) \quad \{\displaystyle \vec{w} \propto (\Sigma_0 + \Sigma_1)^{-1} (\mu_0 - \mu_1)\}$$

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Assumptions The assumptions of discriminant analysis are the same as those for MANOVA. The analysis is quite sensitive to outliers and the size of the smallest group must be larger than the number of predictor variables.

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This means that when $\vec{w}$ is an eigenvector of $\Sigma^{-1}\Sigma_b$ the separation will be equal to the corresponding eigenvalue. If $\Sigma^{-1}\Sigma_b$ is diagonalizable, the variability between features will be contained in the subspace spanned by the eigenvectors corresponding to the $C - 1$ largest eigenvalues (since Σ_b is of rank $C - 1$ at most). These eigenvectors are primarily used in feature reduction, as in PCA. The eigenvectors corresponding to the smaller eigenvalues will tend to be very sensitive to the exact choice of training data, and it is often necessary to use regularisation as described in the next section. If classification is required, instead of dimension reduction, there are a number of alternative techniques available. For instance, the classes may be partitioned, and a standard Fisher discriminant or LDA used to classify each partition. A common example of this is "one against the rest" where the points from one class are put in one group, and

everything else in the other, and then LDA applied. This will result in C classifiers, whose results are combined. Another common method is pairwise classification, where a new classifier is created for each pair of classes (giving $C(C - 1)/2$ classifiers in total), with the individual classifiers combined to produce a final classification.